

Homework 6

Domain & Range: Finding Outputs by Solving Backwards

Functions Unit — Extension

Name: _____

Instructions

Use a **pen** and **show every step**. This is a difficult homework, you must be resourceful. It would be a good idea to draft your answers on a blank sheet of paper before writing your final answer in pen.

For each function you will find the **domain** and the **range**, and for both you must *justify* your answer using the method below. You may use the internet and AI as a **teacher**, but try the problems yourself first.

Reading — Finding the Range by Solving Backwards

Recall from the functions unit: the **domain** is the set of allowed inputs, and the **range** is the set of outputs the function can actually produce. Finding the domain is about avoiding trouble; finding the range takes the clever idea you saw in class.

Domain: avoid the two dangers

For every function in this homework, the domain is all real numbers *except* where the rule breaks. Two things break a rule:

- **Division by zero:** a denominator can never equal 0.
- **Even roots of negatives:** whatever sits inside a square root must be ≥ 0 .

Everywhere else, the input is allowed.

Range: ask “which outputs are reachable?”

Here is the key idea. Pick an *arbitrary* target output y and ask: is there an input x in the domain with $f(x) = y$? To find out, **solve the equation $y = f(x)$ for x** .

- If you can solve it and get a valid x (a real number in the domain), then y is in the range.
- If certain values of y force an impossible x (dividing by zero, or a square root of a negative), then those y are *not* in the range.

*This is exactly the “onto” question from our functions unit: a value y is in the range precisely when it has a **preimage** — an input that maps to it. Solving $y = f(x)$ for x is how you hunt for that preimage.*

Worked example

Find the domain and range of $g(x) = \frac{6x}{x+2}$.

Domain. The denominator $x+2$ equals 0 when $x = -2$, so the domain is all real numbers except -2 .

Range. Set $y = \frac{6x}{x+2}$ and solve for x :

$$\begin{aligned} y(x+2) &= 6x \\ yx + 2y &= 6x \\ yx - 6x &= -2y \\ x(y-6) &= -2y \\ x &= \frac{-2y}{y-6}. \end{aligned}$$

This produces a valid x for every y *except* $y = 6$ (which would divide by zero). So g can output any real number except 6: the range is all real numbers except 6. You can also write this as $(-\infty, 6) \cup (6, \infty)$. You can answer in words or use mathematical symbols, either is fine as long as you are clear.

Now use this same method on each problem: **domain first, then solve $y = f(x)$ for x .**

Problems

For each function, find the domain and range with formal justification.

1. $f(x) = 4x - 7$.

2. $f(x) = \sqrt{2x + 6}$.

3. $g(x) = \frac{5x}{x-3}.$

4. $h(x) = x^2 + 2.$

5. ★ **Challenge.** $p(x) = x^2 - 4x + 7$.

Hint: solving $y = x^2 - 4x + 7$ for x means solving a quadratic equation for x . Think about when that quadratic has a real solution — that is the question that decides the range. (Completing the square is another route.)

Notebook Artifact

N1. If you used the internet or AI, describe how you used it.
